

Laboratory Assistive Hand

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Abstract—We describe the design and experiments with *Laboratory Assistive Hand (LAH)*, a tendon-driven robotic hand that is actuated by a single actuator. The robot hand consists of two fingers with one slider on each. The hand was designed to grasp laboratory equipments and do the titration. The primary focus of this report is to compare the final result with our project description document(PDD), as well as the design of LAH. The experimental results show LAH is able to grasp most objects successfully, and do the titration well.

Keywords—LAH, tendon-driven, underactuated

I. INTRODUCTION

Building a robot assistive hand to grasp and moving objects is a common goal for most end effectors. There are a variety of solutions to approach it. LAH is aimed to manipulating some laboratory tasks in the medical facility and pharmaceutical laboratory so that scientists do not need to conduct experiments and contact the hazardous substances directly. Instead, tasks can be accomplished by a remote control.

One consideration was how to grasp different sizes of glassware/plasticware. To ensure the stability of grasping, LAH performs parallel grasp for most objects as shown in Fig. 1 and envelope grasp if the equipped actuator can provide enough force. With the capability to grasp laboratory glassware and plasticware stably, users can use LAH to complete simple tasks, e.g. pouring solvent from a test tube. Besides grasping ability, The novel design of LAH allows the robot to manipulate the titration easily as shown in Fig. 2; there is not any product that serves our purpose in current market.

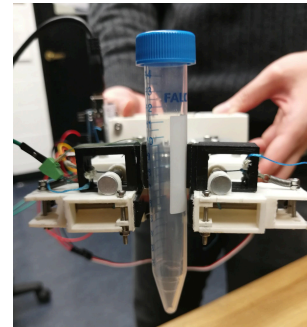


Fig. 1. Parallel grasp

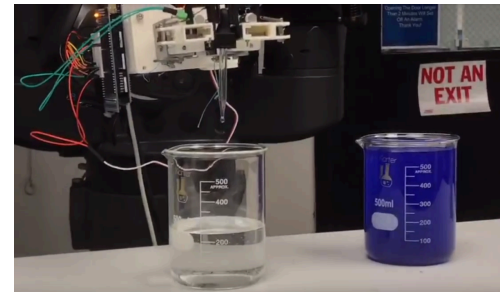


Fig. 2. Titration task

II. HAND DESIGN

A. Basic mechanism

The design of LAH was inspired by the Velo Gripper^[1], using single actuator to drive two tendons to achieve both parallel grasp and enveloping grasps. The pose of LAH is not controlled directly by the actuator which only provides force. It changes along with the object the LAH grasps; extension

and torsion springs will provide the torque and constraints throughout the motion. There are two grasping behaviors we want to achieve; the distal links should be parallel in order to achieve parallel grasp, and the pose of LAH will achieve an envelope grasp if the proximal links are stopped by contact with object.

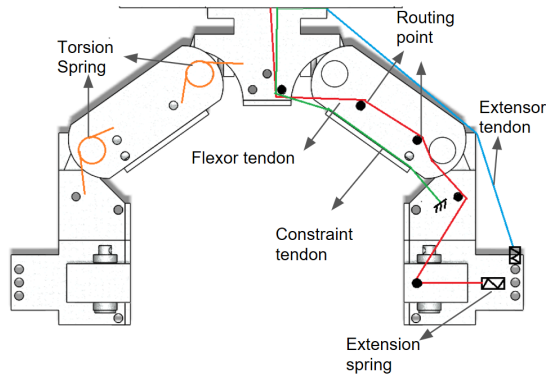


Fig.3 LAH components: flexor tendon (red) with routing points, torsion springs shown at joints (on the opposite finger) for clarity, and extensor tendon (blue) connected to extension spring, constraint tendon (green).

In order to achieve it, there are three tensions on each side to guild the behavior of LAH. As shown in Fig. 3, the extensor tendons, which are connected to a linear spring, provide extension torque so that the distal links of LAH keep parallel during motion. Moreover, constraint tendons are used to keep the distal links parallel at initial state. The flexor tendons are directly connected to the actuator. They apply the force on routing points to produce corresponding motion while the actuator is on.

B. GraspIt! optimization

We use GraspIt![2] to optimize our hand design.

At the beginning, we had our concept design using solidworks and import it in GraspIt! (Fig. 4). GraspIt! is a software which tests how well a designed-hand can grasp an object. GraspIt! will give a quality measure which stands for how firm this grasp is. The initial design had the extensor tendon inside the finger and revolute joint and it gave a quality measure of 0.0742. After testing on GraspIt!, we 3D printed our parts, assembled them and had our first prototype (Fig. 5). However, our first prototype did not work as expected; parallel grasp was not achieved so we redesigned the hand. New design put the extensor tendon outside and then repeated the same process as in Fig. 6 and got a second prototype (Fig. 8)

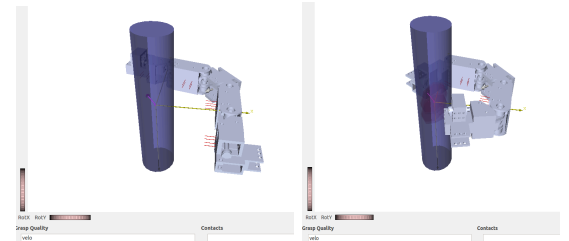


Fig. 4. Initial design imported in GraspIt!

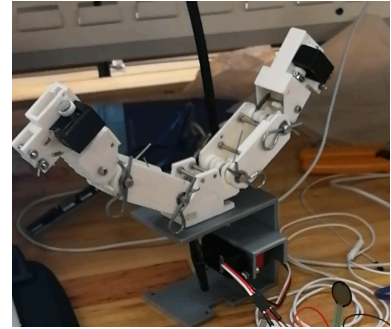


Fig. 5. First prototype after using GraspIt!

Second design had a quality measure of 0.145 and new hand functioned well and gave a parallel grasp. The reason why change of the extensor tendon was new design gave a greater displacement and therefore greater torque.

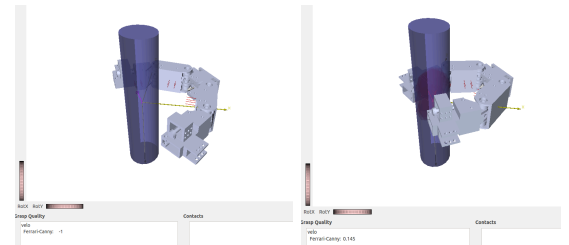


Fig. 6 Second trial in GraspIt!

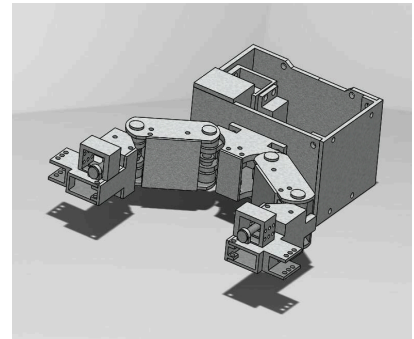


Fig. 7 Second design

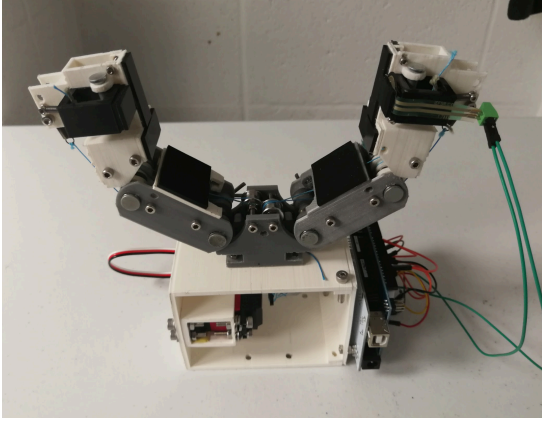


Fig. 8 Second prototype

C. Slider

Other main topic of this project is to do titration, which is accomplished by the novel hand design. Squeezing the dropper is the main problem that we need to solve. To do this manipulating task, it requires very accurate force control. As we have only one actuator, a new supplement was needed. In order to allow the underactuated robot hand to do titration, we reduce the constraints by adding one slider on each finger instead of using a revolute joint. Adding slider is easier to control and gives better safety. The slider is connected to an active tendon, and its motion is constricted by a linear spring, which requires initial tension, so that keep pulling the tendon will let the slider continue to move. An advantage to install a slider on the distal link is that it does not affect the quality of grasping. As long as distal links grasp object, LAH can do the titration easily.

III. MECHATRONICS DESIGN

LAH is underactuated which only uses one actuator to drive the tendon. As mentioned in product description document, we need to use a sensor in our project. The integration of sensor with servo motor to implements force control (Fig. 11). In other word, once the sensor that is installed on the slider touches the objects and readings from a sensor reach the threshold, we can demonstrate that LAH holds object tightly.



Fig. 9 Force sensitive resistor sensor (FSR) and Servo motor



Fig. 10 Baxter

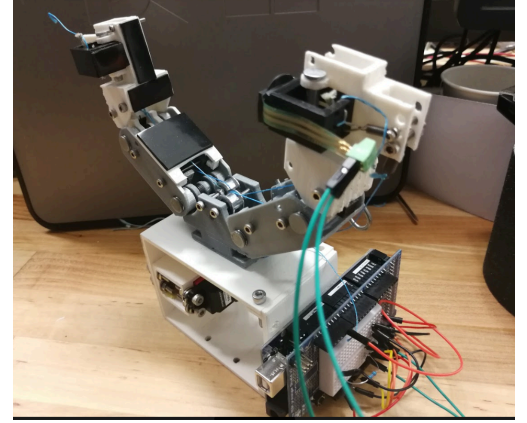


Fig. 11 Sensor, arduino and wiring

We use force sensitive resistor (FSR) (Fig. 9) as sensor. FSR allows user to detect physical pressure and squeezing. It is simple to use and enables feedback control. However, the value read by FSR will vary. Due to the mechanism of FSR, given certain force, contact with different locations of FSR gives different force readings. We mount rubber on FSR to mitigate this type of uncertainty. Moreover, we program an arduino code to mitigate this uncertainty. The reason why we use sensor is to assure that fingers do not squeeze the rubber part of dropper when doing only grasping task. With the use of sensor, it can prevent the actuator from pulling further so that the hand is only grasping without titrating. We select a servo motor (Fig. 9) to supply power as the servo motor is easy to use. Also we use the Baxter (Fig. 10) as a robotic arm and combine ROS systema and arduino to control the motion.

IV. QUANTITATIVE EVALUATION AND RESULT

A. Grasping

According the Table 1, after 20 testing trials, we have firmly grasped a dropper, a test tube, and a beaker with success rate higher than the success criteria mentioned in the

PDD. This testing results indicate our grasping is stable and tight.

Table1. Testing result of grasping objects

	Dropper	Test tube	Beaker
Success rate	80%	90%	80%
Goal	50%	60%	70%

B. Manipulation

Basic goal:

1. Able to lift the test tube from a holder (Fig. 12)
2. Able to control that the liquid in the dropper didn't drop until "squeeze" command was given. (Fig. 2)
3. Able to optimize the hand design using GraspIt!. (Fig. 4 and 6)

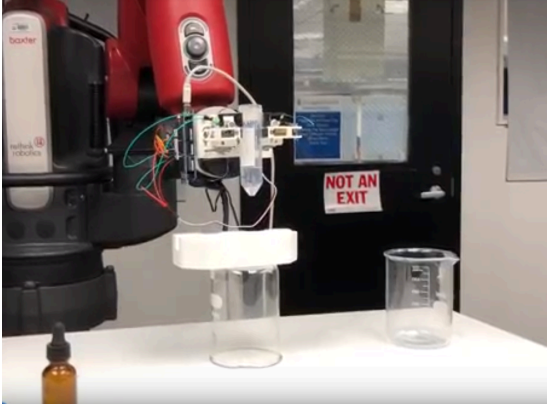


Fig. 12 Lift a test tube from a holder

V. DISCUSSION AND FUTURE WORK

In this report, we designed our underactuated two-finger hand and adopted parallel and enveloping grasp to grasp target items. According to our final test result, it demonstrates that our hand is able to perform firm grasp with most items including our target item: a dropper, test tube, and beaker. Furthermore, we conducted experiments in which we changed the relative location of graspable objects to the hand to see if parallel and enveloping grasps function. As shown in Fig. 13, for each item, it was placed next to the fingertip, next to the second link, and finally right in front of the palm of the hand and "grasp" command was executed. If items were placed next

to the fingertip, parallel grasps were performed. For the test tube, enveloping grasps were successfully performed and grasping process was stopped once the sensor readings reached the specific threshold.

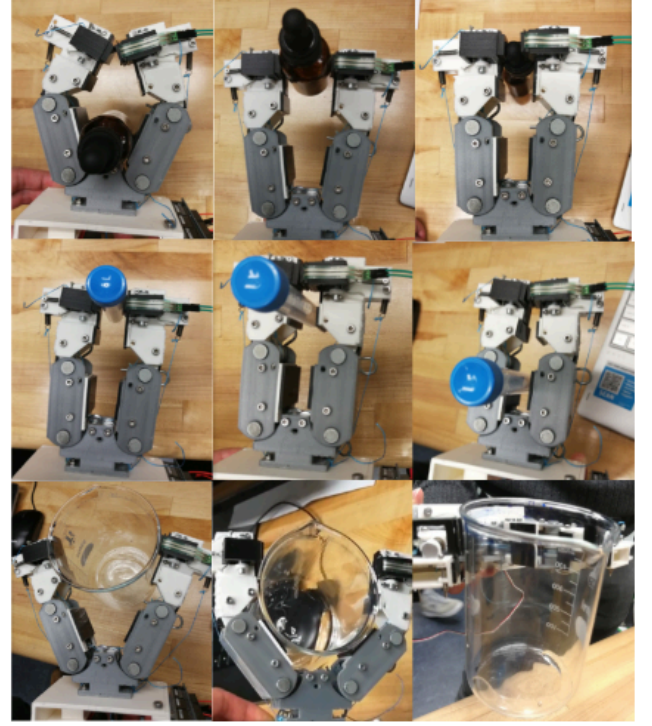


Fig. 13 Grasp target items with different contact positions

One of our main goal is to manipulate the titration process so that it can be applied in medical facilities and chemical laboratories. To see if we were able to control the titration process, we let the Baxter move right above the beaker which was filled with colored water and gave Baxter "squeeze" command so that the dropper sucked up the colored water. Then we moved the dropper right above another beaker with water and squeezed the dropper again. As a result, the whole titration process was smooth and well.

Moreover, lifting the test tube from the holder was also stable and successful. By this point, our LAH has proved capable of grasping most items even with complex shapes. Although grasping is proved to be firm and successful, there is still a lot to be done to improve our hand design and process of grasping and titration.

First of all, redesign of routine points can decrease the friction which is a huge problem as large friction could damage the tendons and larger force is needed to accomplish the same task. Secondly, another additive manufacturing method may be chosen instead of using 3D printing only. The advantage of 3D printing is it can easily achieve complex task. However, 3D printing is time consuming as printing time as well as cleaning time is enormous and more importantly the

precision of printed parts cannot be guaranteed which further worsens the time cost. In addition, more assembly-friendly design is favored as it saves a lot of time. As testing and assembling parts are unavoidable, wasting time for purely assembling purpose is not wise. Therefore, assembly-friendly design could improve the efficiency and stability. Also, more complete and versatile mechatronics design integrated with sensors may greatly reduce the errors occurred during the testing process.

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